OMAR MUKHTAR

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EDUCATION

Bocconi University Milan, Italy
MSc in Data Science and Business Analytics August 2024 – Present
Key Courses: Machine Learning, Algorithms, Optimization, Statistics, Stochastic Processes.
Thesis: Enhancing Out-of-Distribution Detection Using Gaussian Processes.

Bocconi University Milan, Italy
BSc in Economics, Management, and Computer Science September 2021 – July 2024
Key Courses: Computer Science, Machine Learning, Econometrics, Statistics, Game Theory.
Thesis: An Exploration into MCMC Methods for Imbalanced Datasets.

EXPERIENCE

One Tech Milan, Italy
Product Engineering Intern June 2026 – July 2026
Built backend and frontend features for a lending platform, including customer data and pipeline tracking, and prototyped AI tools for SaaS and FinTech clients.

- Developed and built core pipeline logic for a loan conversion workflow on a BaaS lending platform built for major European lenders.
- Linked customer contact details, platform chats and uploaded documents to a single customer record across several sources, keeping profiles up to date for lender teams.
- Prototyped an AI shopping assistant for Shopify storefronts, using MCP servers to call store APIs directly and answer customer product questions, support navigation, and provide contextual shopping guidance.
- Traced and diagnosed errors during testing and development using SQL, accelerating debugging and improving release reliability.

Valueize Milan, Italy
Junior Product Intern November 2024 – April 2025
Contributed to the launch of a FinTech platform offering automated financial assistance.

- Validated an LLM-based NLP model identifying 35,000+ data points, leading to an overall model accuracy of 97.5%.
- Designed and supported the development of an internal quality control and assurance platform to be used for all future QA&C operations, focusing on minimizing human error and involvement, increasing automation, and improving efficiency.
- Supported the development of API integration with leading E-commerce, accounting, and analytics software, identifying relevant endpoints and designing efficient data transformation logic, optimizing interoperability.
- Conducted a comprehensive competitive analysis to provide optimal user experience with software integration in anticipation of the product launch.

PROJECTS

Information Asymmetry & Divergence of Opinion Modelling
Studied how differences in investor beliefs about a stock can predict its future performance.

- Engineered a machine learning model that measures how differently investors interpret the same information, using over 14 million data points across 500+ S&P 500 companies.
- Delivered a trading strategy based on this signal that returned 22.3% annually over 10 years, beating a standard market benchmark by 7.4 percentage points.

EcoEyeSpy – Illegal Landfill Detection from Aerial Imagery
Built a computer-vision pipeline that screens aerial and satellite images to detect and map illegal waste dumping sites.

- Trained an image classifier that catches 98% of illegal waste sites under severe class imbalance.
- Developed an object detector that locates individual waste piles with 0.75 average IoU on correct detections, using only 1.4% labeled data (166 of 11,700+ images).

Ongoing Projects

- **Rare-Event Prediction in Fantasy Football**: Building a prediction system for Fantasy Premier League with a focus on flagging the top 10% of predicted scores each week, currently capturing 72% of such scores.

LANGUAGE & IT SKILLS

Native Languages: English, Arabic
Other Languages: Italian
Programming and IT Skills: Python, SQL, R, Git, Microsoft Office, AWS
Portfolio Website: https://omarn-mukhtar.github.io/
GitHub: https://github.com/OmarN-Mukhtar